

Sahoj Patkar

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[LinkedIn Profile](#) | Portfolio Website

EDUCATION

J.P. Stevens High School | Edison, NJ

High School Diploma Candidate | Graduation Year: 2029

TECHNICAL SKILLS

- **Programming:** Java, Kotlin, Python, C++
- **Robotics & Hardware:** Autonomous Navigation, GPS Waypoint Tracking, 3-DOF Arm Kinematics, Flight Controllers, PCB Design, 3D Printing (FDM), Sensor Integration, Soldering
- **Software & Tools:** Android Studio, Arduino IDE, GitHub, OpenCV, PyTorch, MediaPipe, LeRobot

ENGINEERING EXPERIENCE

SkyRelay | Hardware Engineering Intern | Summer 2026 – Present

- Utilize KiCad tools to design, route, and prototype multi-layer Printed Circuit Boards (PCBs) for embedded hardware systems.
- Manage Bill of Materials (BOM) and component selection to optimize power distribution efficiency and signal integrity.

Rutgers University STAR Sloshing Platform for In-Orbit Controller Experimentation Sat Team | Systems & Software Engineering Subteam Member

Summer 2026 – Present

- Developing NASA core Flight System (**cFS**) applications on the On-Board Computer (NANOMIND SDR MK3 / Zynq 7000) to interface with the Raspberry Pi payload computer via **RS-422 full-duplex serial link**.
- Utilizing the **NOS3** (NASA Operational Simulator for Small Satellites) virtual environment to model, test, and validate spacecraft flight software protocols.
- Authored and structured **Requirements Verification Matrices (RVM)** templates for University Nanosatellite Program (UNP) compliance, including UNP inspection sign-off checklists and flight fastener torque logs (per MSFC-STD-486B).
- Implementing protocol-level packet structures and byte-synchronization handling (using CCSDS Space Packet Standards and **0x1ACFFC1D** sync words) based on the Payload-OBC Interface Control Document (ICD).

Google Developer Program | Member | Fall 2025 – Present

- Collaborate on community projects focused on cloud infrastructure, Android architecture, and emerging AI tools.

PROJECTS & PORTFOLIO

Autonomous Mail-Retrieval Drone | Lead Developer | Expected Completion: Fall 2027

- Engineered an autonomous drone utilizing GPS-guided waypoints for precision navigation and localized targeting.
- Designed and integrated a custom 3-DOF robotic arm calibrated for high-precision gripping and secure object retrieval.
- Programmed custom stabilization algorithms and automated Smart Return-to-Home (SRTH) radio telemetry failsafes.

Edge-AI Sign Language Translator | Developer | Spring 2026

- Architected a real-time ASL translator using Python and MediaPipe to map 21 hand-joint coordinates in 3D space.
- Optimized inference engine for a Raspberry Pi 3 using heuristic geometric logic to achieve low-latency processing.
- Integrated a multi-threaded text-to-speech engine with a noise-filtering frame-buffer algorithm to reach 95%+ accuracy.

HuggingFace AI Robotic Arm (LeRobot S0101) | ML Developer | Fall 2025-Present

- Utilized Multiview OpenCV and Python to implement real-time object detection, classification, and spatial tracking.
- Trained an Action Chunking Transformer (ACT) policy using demonstration data collected with LeRobot to enable autonomous manipulation on the physical S0101 robotic arm.

Co-Founded XChange | Co-Founder & Lead Developer | Winter 2025

- Architected and deployed a secure mobile application using Kotlin and Java to facilitate peer-to-peer local item exchanges.

LEADERSHIP & ACTIVITIES

FTC Robotics Team | Team Captain, Build & Code Lead | Fall 2024 – Present

- Developed an autonomous navigation system using dead-reckoning odometry and path planning in Android Studio.

- Designed and 3D-printed custom structural chassis components; established team Git/GitHub workflows.

Athletics & Arts: Varsity/Club Swim Team (State Champion, 4 consecutive years); CamelBack Ski Race Team (8 years); Carnegie Hall Vocalist (Selective Choir).

AWARDS & ACHIEVEMENTS

- **FTC Autonomous Pathing Hackathon:** 1st Place Winner.
- **FTC Rookie Year Design Award:** Awarded for excellence in mechanical design and custom 3D fabrication.
- **Congressional App Challenge:** Official mobile application entry and competition participant.